

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT : DATA STRUCTURE

Practical No : 10

Aim : Write a program to create a basic Hash Table for insertion, deletion, traversal operations(assume that there are no collisions).

CODE :

```
print("Om Wala S119")
```

```
SIZE = 10
```

```
hash_table = [None] * SIZE
```

```
def hash_function(key):  
    return key % SIZE
```

```
def insert(key, value):  
    index = hash_function(key)  
  
    if hash_table[index] is None:  
        hash_table[index] = (key, value)  
        print("Inserted:", key, "->", value)  
    else:  
        print("Index already occupied. Collision occurred.")
```

```
def delete(key):  
    index = hash_function(key)  
  
    if hash_table[index] is not None and hash_table[index][0] == key:  
        hash_table[index] = None  
        print("Deleted key:", key)  
    else:  
        print("Key not found.")
```

```
def traverse():  
    print("\nHash Table:")  
    for i in range(SIZE):  
        if hash_table[i] is not None:  
            print(i, ":", hash_table[i][0], "->", hash_table[i][1])  
        else:  
            print(i, ": Empty")
```

```
while True:  
    print("\n--- Hash Table Menu ---")  
    print("1. Insert")
```

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```
print("2. Delete")
print("3. Traverse")
print("4. Exit")

choice = int(input("Enter your choice: "))

if choice == 1:
    key = int(input("Enter key: "))
    value = input("Enter value: ")
    insert(key, value)

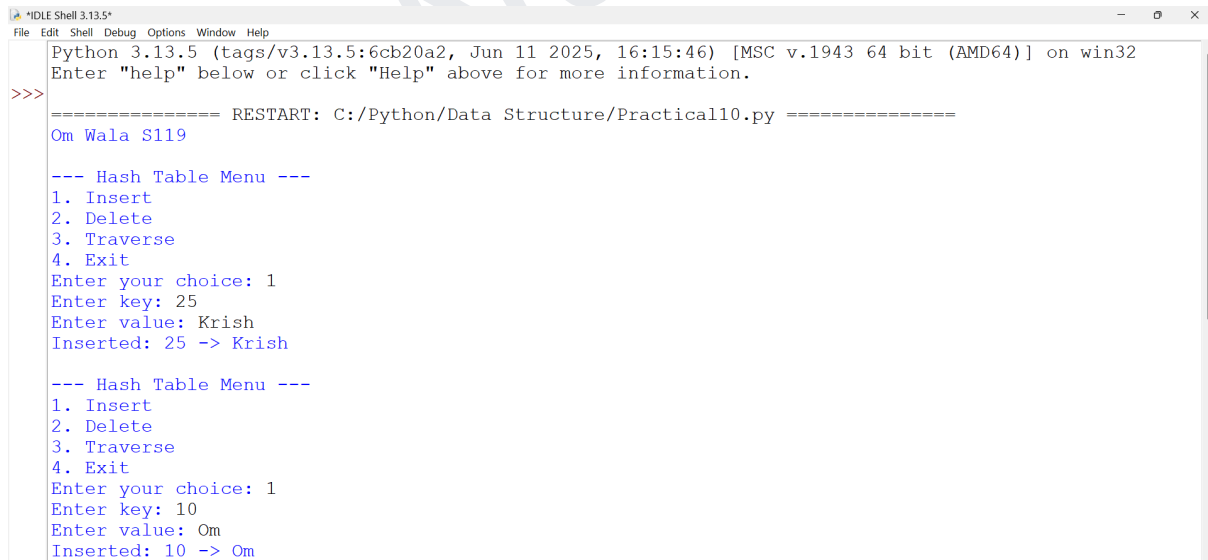
elif choice == 2:
    key = int(input("Enter key to delete: "))
    delete(key)

elif choice == 3:
    traverse()

elif choice == 4:
    print("Program ended.")
    break

else:
    print("Invalid choice.")
```

OUTPUT :



```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Python/Data Structure/Practical10.py =====
Om Wala S119

--- Hash Table Menu ---
1. Insert
2. Delete
3. Traverse
4. Exit
Enter your choice: 1
Enter key: 25
Enter value: Krish
Inserted: 25 -> Krish

--- Hash Table Menu ---
1. Insert
2. Delete
3. Traverse
4. Exit
Enter your choice: 1
Enter key: 10
Enter value: Om
Inserted: 10 -> Om
```

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```
*IDLE Shell 3.13.5*
File Edit Shell Debug Options Window Help

--- Hash Table Menu ---
1. Insert
2. Delete
3. Traverse
4. Exit
Enter your choice: 3

Hash Table:
0 : 10 -> Om
1 : Empty
2 : Empty
3 : Empty
4 : Empty
5 : 25 -> Krish
6 : Empty
7 : Empty
8 : Empty
9 : Empty

--- Hash Table Menu ---
1. Insert
2. Delete
3. Traverse
4. Exit
Enter your choice: 2
Enter key to delete: 25
Deleted key: 25

--- Hash Table Menu ---
1. Insert
2. Delete
3. Traverse
4. Exit
Enter your choice: 3

Hash Table:
0 : 10 -> Om
1 : Empty
2 : Empty
3 : Empty
4 : Empty
5 : Empty
6 : Empty
7 : Empty
8 : Empty
9 : Empty

--- Hash Table Menu ---
1. Insert
2. Delete
3. Traverse
4. Exit
Enter your choice: 4
Program ended.
>>>
```

CODE GUI :

```
import tkinter as tk
from tkinter import messagebox
```

```
SIZE = 10
hash_table = [None] * SIZE
```

```
def hash_function(key):
    return key % SIZE
```

```
def insert():
    try:
        key = int(key_entry.get())
        value = value_entry.get()
```

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```
if value == "":
    messagebox.showwarning("Warning", "Please enter a value.")
    return

index = hash_function(key)

if hash_table[index] is None:
    hash_table[index] = (key, value)
    messagebox.showinfo("Success", "Key inserted successfully.")
else:
    messagebox.showerror("Error", "Collision occurred. Index already occupied.")

display_table()

except ValueError:
    messagebox.showerror("Error", "Key must be an integer.")

def delete():
    try:
        key = int(key_entry.get())
        index = hash_function(key)

        if hash_table[index] is not None and hash_table[index][0] == key:
            hash_table[index] = None
            messagebox.showinfo("Success", "Key deleted successfully.")
        else:
            messagebox.showerror("Error", "Key not found.")

        display_table()

    except ValueError:
        messagebox.showerror("Error", "Key must be an integer.")

def display_table():
    output.delete("1.0", tk.END)

    output.insert(tk.END, "INDEX\tKEY\tVALUE\n")
    output.insert(tk.END, "-" * 35 + "\n")

    for i in range(SIZE):
        if hash_table[i] is not None:
            key, value = hash_table[i]
            output.insert(tk.END, f"{i}\t{key}\t{value}\n")
        else:
            output.insert(tk.END, f"{i}\t-\t-\n")

def clear_fields():

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```

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```
key_entry.delete(0, tk.END)  
value_entry.delete(0, tk.END)
```

```
root = tk.Tk()  
root.title("Hash Table - Om Wala S119")  
root.geometry("650x600")  
root.configure(bg="#F4F6F8")
```

```
title = tk.Label(  
    root,  
    text="Hash Table",  
    font=("Segoe UI", 24, "bold"),  
    bg="#F4F6F8",  
    fg="#333333"  
)  
title.pack(pady=15)
```

```
student = tk.Label(  
    root,  
    text="Om Wala S119",  
    font=("Segoe UI", 12, "bold"),  
    bg="#F4F6F8",  
    fg="#4A90E2"  
)  
student.pack()
```

```
frame = tk.Frame(root, bg="white", padx=20, pady=20)  
frame.pack(pady=20)
```

```
tk.Label(  
    frame,  
    text="Key:",  
    font=("Segoe UI", 12),  
    bg="white"  
)  
.grid(row=0, column=0, padx=10, pady=10)
```

```
key_entry = tk.Entry(  
    frame,  
    font=("Segoe UI", 12),  
    width=25  
)  
key_entry.grid(row=0, column=1, padx=10, pady=10)
```

```
tk.Label(  
    frame,  
    text="Value:",  
    font=("Segoe UI", 12),
```

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```
        bg="white"
    ).grid(row=1, column=0, padx=10, pady=10)

    value_entry = tk.Entry(
        frame,
        font=("Segoe UI", 12),
        width=25
    )
    value_entry.grid(row=1, column=1, padx=10, pady=10)

    insert_button = tk.Button(
        frame,
        text="Insert",
        font=("Segoe UI", 11, "bold"),
        bg="#4A90E2",
        fg="white",
        width=12,
        command=insert
    )
    insert_button.grid(row=2, column=0, padx=5, pady=15)

    delete_button = tk.Button(
        frame,
        text="Delete",
        font=("Segoe UI", 11, "bold"),
        bg="#e74c3c",
        fg="white",
        width=12,
        command=delete
    )
    delete_button.grid(row=2, column=1, padx=5, pady=15)

    clear_button = tk.Button(
        frame,
        text="Clear",
        font=("Segoe UI", 11, "bold"),
        bg="#50C878",
        fg="white",
        width=12,
        command=clear_fields
    )
    clear_button.grid(row=3, column=0, columnspan=2, pady=5)

    tk.Label(
        root,
        text="Hash Table Traversal",
        font=("Segoe UI", 15, "bold"),
        bg="#F4F6F8",
```

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```
fg="#333333"  
) .pack(pady=5)  
  
output = tk.Text(  
    root,  
    font=("Consolas", 12),  
    width=55,  
    height=13  
)  
output.pack(pady=10)  
  
display_table()  
  
root.mainloop()
```

OUTPUT :

Hash Table

Om Wala S119

Key: 10
Value: Om Wala

Hash Table Traversal

INDEX	KEY	VALUE
0	10	Om Wala
1	-	-
2	-	-
3	-	-
4	-	-
5	25	Shreyash Kadam
6	36	Krish Gupta
7	-	-
8	-	-
9	9	Deepak Sahani

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Hash Table - Om Wala S119

Hash Table

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Key: 10

Value: Om Wala

InsertDeleteClear

Error

Collision occurred. Index already occupied.

OK

INDEX	KEY	VALUE
0	1	-
1	-	-
2	-	-
3	-	-
4	-	-
5	25	Shreyash Kadam
6	36	Krish Gupta
7	-	-
8	-	-
9	9	Deepak Sahani

Hash Table - Om Wala S119

Hash Table

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Key: 9

Value: Deepak Sahani

InsertDeleteClear

Success

Key deleted successfully.

OK

INDEX	KEY	VALUE
0	10	-
1	-	-
2	-	-
3	-	-
4	-	-
5	25	Shreyash Kadam
6	36	Krish Gupta
7	-	-
8	-	-
9	9	Deepak Sahani

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Hash Table - Om Wala S119

Hash Table

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Key:

Value:

Insert

Delete

Clear

Hash Table Traversal

INDEX	KEY	VALUE
0	10	Om Wala
1	-	-
2	-	-
3	-	-
4	-	-
5	25	Shreyash Kadam
6	36	Krish Gupta
7	-	-
8	-	-
9	-	-

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